TÉCNICO
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DE LISBOA

EXERCISE 1.5 · SHIP AND OCEAN SYSTEMS DESIGN

Displacement equation: review and application of fixed-point iteration and Newton–Raphson

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Individual practice for D3 · not graded
Papanikolaou (2026) §2.13, Appendix C

Questions

1. Write the equilibrium equation, buoyancy = weight. Identify the known input and the unknown, and justify Δ_0 = DWT.
2. Using the hull proportions, express L, B, T and D as functions of Δ.
3. Show that the equipment numeral and the steel weight are powers of Δ, and find their coefficients.
4. Write the fixed-point iteration $\Delta_{k+1} = g(\Delta_k)$, with all coefficients evaluated.
5. Without iterating, prove that the iteration converges for any Δ ≥ DWT, and interpret $g'(\Delta)$.
6. Iterate from Δ_0 = DWT until the change is below 1 t. Give Δ*, the main dimensions and the weight groups.
7. Does the set-up matter? Compare Form A with Form B, which starts from the machinery.
8. Solve the equilibrium with Newton–Raphson from Δ_0 = DWT, and compare the iterations and the convergence rate.

Review of numerical methods

Theory · Fixed-point iteration (successive substitution, Picard)

$f(x) = 0 \iff x = g(x) \qquad x_{k+1} = g(x_k)$

$|g'(x)| < 1 \text{ near } x^* \;\Rightarrow\; e_{k+1} \approx g'(x^*)\, e_k$

Needs only g. Error × |g′(x*)| per pass: **linear**.

Theory · Newton–Raphson (tangent method)

$f(x) = 0 \qquad x_{k+1} = x_k - \dfrac{f(x_k)}{f'(x_k)} \qquad f'(x^*) \neq 0 \;\Rightarrow\; e_{k+1} \approx \dfrac{f''(x^*)}{2f'(x^*)}\, e_k^2$

Needs f and f′, a start close enough. Error squared: **quadratic**.

Example $\Delta = \underbrace{10\,000}_{\text{DWT}} + \underbrace{150\sqrt{\Delta}}_{\text{lightship}}$ Start at Δ_0 = DWT = 10 000 t · stop when the change is **below 0.01 t**

To compose the fixed-point iteration

1 · f = 0 $f(\Delta) = \Delta - 10\,000 - 150\sqrt{\Delta} = 0 \iff \Delta = 10\,000 + 150\sqrt{\Delta}$

2 · g, g′ $g(\Delta) = 10\,000 + 150\sqrt{\Delta} \quad g'(\Delta) = \dfrac{75}{\sqrt{\Delta}} \le 0.75 < 1$ ✓

3 · iterate $\Delta_{k+1} = 10\,000 + 150\sqrt{\Delta_k}$

To compose Newton–Raphson

1 · f, f′ $f(\Delta) = \Delta - 10\,000 - 150\sqrt{\Delta} \quad f'(\Delta) = 1 - \dfrac{75}{\sqrt{\Delta}} \ge 0.25 \neq 0$ ✓

2 · iterate $\Delta_{k+1} = \Delta_k - \dfrac{\Delta_k - 10\,000 - 150\sqrt{\Delta_k}}{1 - 75/\sqrt{\Delta_k}}$

Same root, 40 000 t: 17 evaluations of g against 5 Newton steps

Example $\Delta = \underbrace{10\,000}_{\text{DWT}} + \underbrace{150\sqrt{\Delta}}_{\text{lightship}}$ Start at Δ_0 = DWT = 10 000 t · count the evaluations of g until the change is **below 0.01 t**

k	Δ_k	$150\sqrt{\Delta_k}$	Δ_{k+1}	change	ratio
0	10 000.0	15 000.0	25 000.0	+15 000.0	
1	25 000.0	23 717.1	33 717.1	+8 717.1	0.581
2	33 717.1	27 543.3	37 543.3	+3 826.2	0.439
3	37 543.3	29 064.1	39 064.1	+1 520.8	0.397
⋮	⋮	⋮	⋮	⋮	⋮
14	39 999.95	29 999.98	39 999.98	+0.032	0.375
15	39 999.98	29 999.99	39 999.99	+0.012	0.375
16	39 999.99	30 000.00	40 000.00	+0.005 ✓	0.375

Fixed-point iteration: 17 evaluations of g (k = 0 to 16). Change × 0.375 = g′(Δ*) per pass: linear.

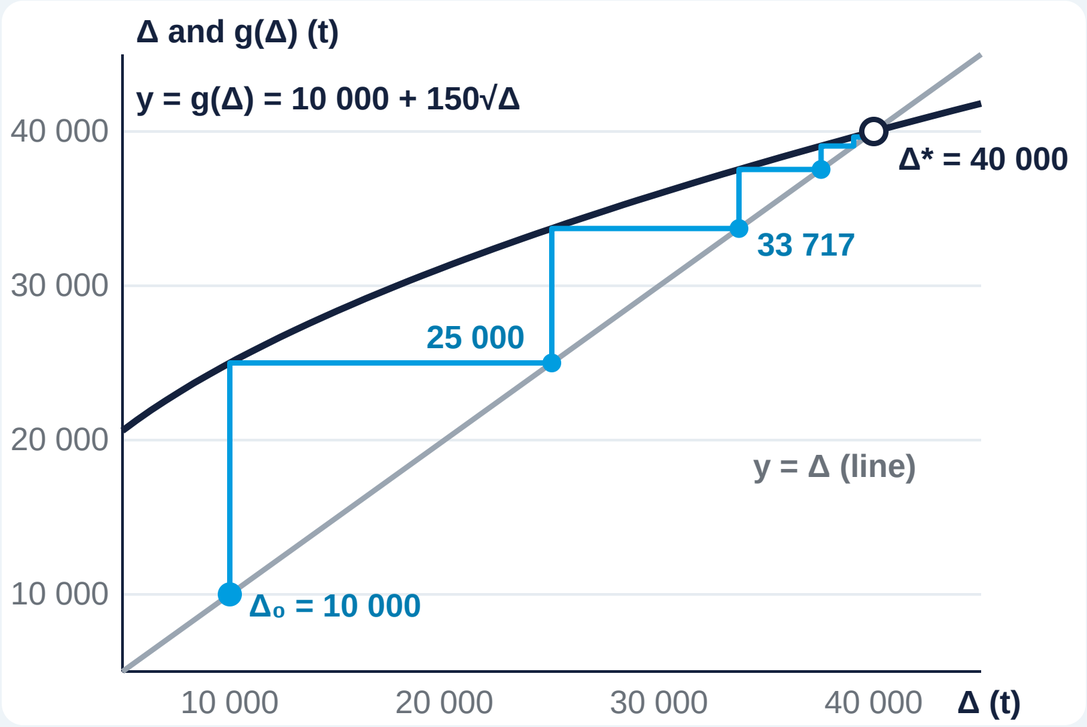
k	Δ_k	$f(\Delta_k)$	$f'(\Delta_k)$	Δ_{k+1}	change
0	10 000.0	−15 000.0	0.250	70 000.0	+60 000.0
1	70 000.0	+20 313.7	0.717	41 649.7	−28 350.3
2	41 649.7	+1 037.3	0.633	40 009.7	−1 640.0
3	40 009.7	+6.05	0.625	40 000.0	−9.68
4	40 000.0	+0.0002	0.625	40 000.0	−0.0004 ✓

k = 0: 15 000 t short, slope only 0.25: grow by 15 000 / 0.25 = 60 000 t, far past Δ*.

Newton–Raphson: 5 steps (k = 0 to 4), each one evaluation of g and g′. Once close, error 1 650 → 9.7 → 0.0004 t: squared each step, quadratic.

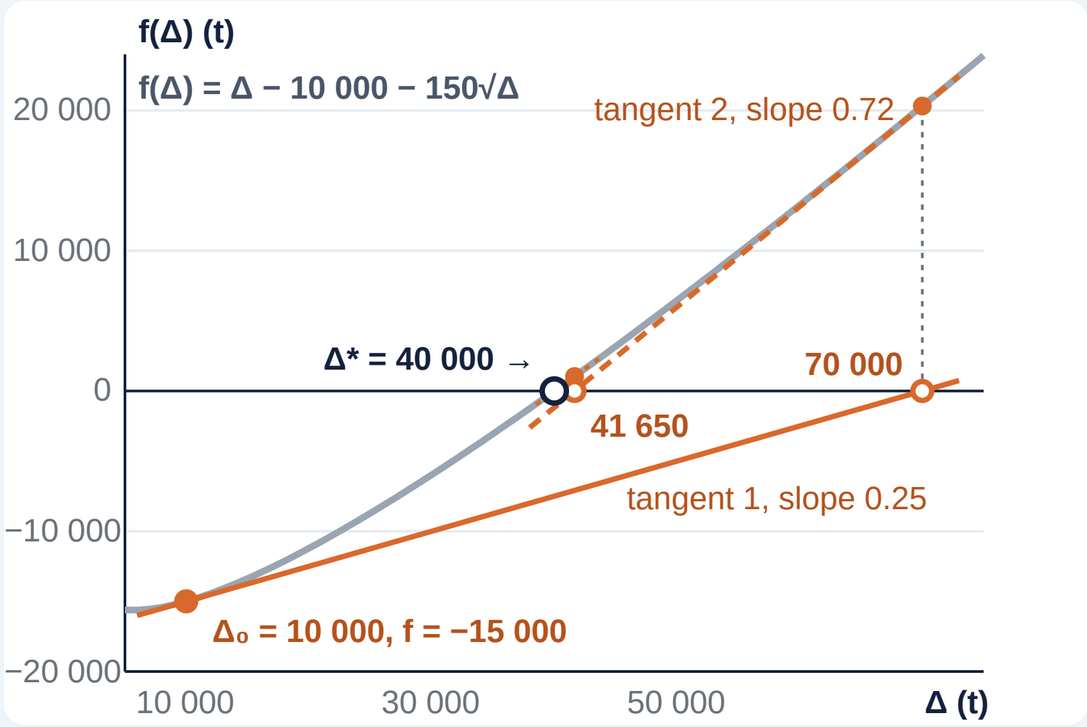
Fixed point climbs a staircase; Newton follows tangents

Fixed-point iteration · Δ = g(Δ)



Δ = g(Δ) = DWT + WLS(Δ): the equilibrium is “buoyancy = weight computed from the current size”.

Newton–Raphson · f(Δ) = 0



f(Δ) = buoyancy − weight: the equilibrium is “excess buoyancy = 0”.

Sizing the reference tanker: 37 600 t deadweight

Objective: determine the displacement Δ and the main dimensions using simplified model equations and an iterative solution. The owner's brief (0.6) fixes the deadweight and the service speed; the proportions and the block coefficient come from Exercise 1.4. The weight of the ship is assumed to break down as:

$$W = \mathrm{DWT} + \underbrace{W_{\mathrm{st}} + W_{\mathrm{o}} + W_{\mathrm{m}}}_{W_{\mathrm{LS}}}$$

Steel	$W_{\mathrm{st}} = K\,E^{1.36} \quad E = L\,(B+T) + 0.85\,L\,(D-T)$
Outfit	$W_{\mathrm{o}} = c_{\mathrm{o}}\,\Delta$
Machinery	$W_{\mathrm{m}} = k_{\mathrm{m}}\,P_{\mathrm{B}} \quad P_{\mathrm{B}} = \dfrac{\Delta^{2/3}\,V^{3}}{C_{\mathrm{adm}}}$

Item	Symbol	Value
Deadweight	DWT	37 600 t
Service speed	V	14.5 kn
Seawater density	ρ	1.025 t/m³
Block coefficient (Ex 1.4)	C_{B}	0.757
Length / beam (Ex 1.4)	L/B	5.92
Beam / draught (Ex 1.4)	B/T	2.69
Depth / draught (Ex 1.4)	D/T	1.36
Steel coefficient (tankers)	K	0.032
Outfit coefficient	c_{o}	0.025
Machinery weight	k_{m}	0.10 t/kW
Admiralty coefficient	C_{adm}	450